Momentum-Space QCD Sum-Rule Setup for the $B_c(1P)$ Axial-Vector Mixing Angle

1 Purpose and conventions

We study the mixing between the two axial-vector $B_c(1P)$ basis states associated with the $^3P_1$ and $^1P_1$ structures. The current implementation follows the Aliev et al. mixing-angle method, but the OPE is evaluated in momentum space because both quark lines are heavy.

This note records the calculation implemented in `BcMixingMomentum.wl`. It is written so that every convention and intermediate object can be checked independently before using the numerical angle in a paper.

The quark content convention is

$$B_c^+ = \bar{b}c. \tag{1}$$

The two interpolating currents are taken as

$$J_\mu^A = \bar{b}\,\gamma_\mu\gamma_5\,c, \tag{2}$$

$$J_\mu^B = i\,\bar{b}\,\sigma_{\mu\alpha}\frac{p^\alpha}{m_b+m_c}\gamma_5\,c, \tag{3}$$

where

$$\sigma_{\mu\nu} = \frac{i}{2}[\gamma_\mu, \gamma_\nu]. \tag{4}$$

The explicit factor of i in J_μ^B is retained. This is important because FeynCalc's `DiracSigma` already contains the conventional $i/2$ commutator.

The factor $1/(m_b+m_c)$ is also essential. Since $[\bar{Q}Q] = 3$ and $[p^\alpha] = 1$, the unnormalized tensor current $i\bar{b}\sigma_{\mu\alpha}p^\alpha\gamma_5 c$ has mass dimension 4, while J_μ^A has mass dimension 3. The normalized current above makes both currents dimension 3.

If the opposite flavor convention is used, for example $B_c^- = \bar{c}b$, the sign of the quoted angle can change. The absolute convention must therefore be kept fixed when comparing to other papers.

2 Mass-dimension check

The quark field has dimension 3/2, while p^α, m_b, and m_c have dimension 1. Therefore

$$[J_\mu^A] = 3, \qquad [J_\mu^B] = 3 \tag{5}$$

only after the tensor current is normalized by $m_b + m_c$.

For currents with dimensions d_i and d_j,

$$[\Pi_{\mu\nu}^{ij}] = d_i + d_j - 4, \tag{6}$$

because of the d^4x integration in the two-point function. Thus the normalized basis gives

$$[\Pi_1^{AA}] = [\Pi_1^{AB}] = [\Pi_1^{BB}] = 2. \tag{7}$$

The spectral densities have the same mass dimension as the invariant correlator,

$$[\rho^{AA}] = [\rho^{AB}] = [\rho^{BB}] = 2. \tag{8}$$

In the present code the Borel moment is evaluated as $\int ds\, e^{-s/M^2}\rho(s)$, so all three numerical moments carry one common extra factor of $[ds] = 2$. This common convention does not affect the mixing angle.

If the tensor current were left unnormalized, the dimensions would instead be

$$[\rho^{AA}],\ [\rho^{AB}],\ [\rho^{BB}] = 2,\ 3,\ 4, \tag{9}$$

and the matrix entries in the angle formula would not be dimensionally compatible.

3 Implementation map

The main code objects are:

- `Correlator[channel, order]`: $\Pi_1^{ij}(p^2)$ after FeynCalc Dirac algebra and spin projection.
- `SpectralDensity[channel, ‘‘pert’’, s]`: $\rho^{ij}_{\rm pert}(s)$.
- `NumericBorelPi[channel, order, M2, s0]`: $\Pi^{ij}_{\rm order}(M^2, s_0)$.
- `NumericMixingAngleDegrees[M2, s0, ‘‘total’’]`: $\theta(M^2, s_0)$ from $\Pi^{ij}_{\rm pert} + \Pi^{ij}_{G^2} + \Pi^{ij}_{G^3}$.

The implemented OPE orders are

$$\text{pert}, \qquad G^2, \qquad \text{pertG2} = \text{pert} + G^2, \qquad G^3, \qquad \text{total} = \text{pert} + G^2 + G^3. \tag{10}$$

The implemented dimension-6 triple-gluon condensate $\langle g_s^3 G^3\rangle$ is the standard vacuum-averaged single-line propagator term,

$$\Pi^{ij}_{G^3,\rm single} = \Pi^{ij}\left[S_c^{(G^3)} S_b^{(0)}\right] + \Pi^{ij}\left[S_c^{(0)} S_b^{(G^3)}\right]. \tag{11}$$

This is the convention used in both the momentum-space calculation and the coordinate-space OPE calculation. The complete dimension–6 basis generated by multiplying two background-field heavy-quark propagators also contains cross-line open-field terms,

$$\Pi^{ij}_{G^3,\rm cross} = \Pi^{ij}\left[S_c^{(GG,\rm open)} S_b^{(G,\rm open)}\right] + \Pi^{ij}\left[S_c^{(G,\rm open)} S_b^{(GG,\rm open)}\right]. \tag{12}$$

These terms are not contained in $\Pi^{ij}_{G^3,\rm single}$. They require the explicitly open two-gluon heavy-quark propagator $S_Q^{(GG,\rm open)}$, i.e. the unaveraged object before replacing two background gluons by $\langle g_s^2 G^2\rangle$. The audit helper `BcMixingDimension6Complete.wl` records this missing sector and should be used before any result is labelled $G^3_{\rm complete}$.

4 How to reproduce the calculation while reading this note

This section is a practical roadmap for a new student. The goal is not to evaluate the final number immediately, but to check one layer at a time. A useful order is:

1. Check the current definitions and mass dimensions. In particular, verify that the factor $p^\alpha/(m_b + m_c)$ in J_μ^B makes the AA, AB, and BB correlators dimensionally compatible.
2. Reproduce the Wick contractions for $\Pi^{AA}_{\mu\nu}$, $\Pi^{AB}_{\mu\nu}$, and $\Pi^{BB}_{\mu\nu}$. At this stage do not calculate any integrals; only check the order of gamma matrices and signs.

3. Apply the spin–1 projector
$$P_{(1)}^{\mu\nu} = \frac{1}{3}\left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}\right)$$
and verify that a scalar invariant Π_1^{ij} is obtained.

4. For the perturbative term, put the two internal quarks on shell and verify the three numerator functions N^{AA}, N^{AB}, and N^{BB} given below. This is the most important algebraic check.

5. Verify the phase-space factor
$$\frac{N_c}{16\pi^2}\frac{\sqrt{\lambda(s, m_b, m_c)}}{s}$$
and the physical threshold $s_{\rm th} = (m_b + m_c)^2$.

6. Perform the Borel integral
$$\Pi^{ij}(M^2, s_0) = \int_{s_{\rm th}}^{s_0} ds\, e^{-s/M^2} \rho^{ij}(s)$$
and only then insert the three Borel moments into the mixing-angle formula.

7. Add condensates only after the perturbative result is stable. For G^2, keep the local single-line terms and the cross-line open-field term conceptually separate; otherwise it is easy to double count or miss a contribution.

The Mathematica workflow mirrors these steps:

Hand calculation object	Mathematica check
current dimensions	`MixingMassDimensionReport[]`
projected trace	`Correlator[channel,"pert"]`
perturbative numerator	`PerturbativeNumerator[channel,s]`
spectral density	`PerturbativeSpectralDensity[channel,s]`
Borel moment	`NumericBorelPi[channel,order,M2,s0]`
mixing angle	`NumericMixingAngleDegrees[M2,s0,order]`
coordinate trace	`CoordinateTraceKernel[channel,order]`
coordinate angle	`CoordinateNumericMixingAngleDegrees[M2,s0,order]`

When a new term is added, the recommended debugging procedure is:

1. compute one channel, usually AA, at one point;

2. check that the result is real;

3. check its mass dimension;

4. check its sign and size against the corresponding momentum-space expression, if available;

5. only then scan over M^2 and s_0.

5 Correlators and spin-1 projection

Define
$$\Pi_{\mu\nu}^{ij}(p) = i \int d^4x\, e^{ip\cdot x} \langle 0|T\{J_\mu^i(x) J_\nu^{j\dagger}(0)\}|0\rangle, \qquad i, j \in \{A, B\}. \tag{13}$$

The invariant amplitude used for the axial-vector spin-1 state is extracted with

$$P^{\mu\nu}_{(1)} = \frac{1}{3}\left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}\right), \tag{14}$$

so that

$$\Pi_1^{ij}(p^2) = P^{\mu\nu}_{(1)}\Pi^{ij}_{\mu\nu}(p). \tag{15}$$

The mixing angle is then obtained from

$$\tan 2\theta = -\frac{2\Pi_1^{AB}}{\Pi_1^{AA} - \Pi_1^{BB}}. \tag{16}$$

Numerically we use the quadrant-safe definition

$$\theta_{\rm raw} = \frac{1}{2}\,\mathrm{atan2}\left(-2\Pi_1^{AB}, \Pi_1^{AA} - \Pi_1^{BB}\right), \tag{17}$$

and quote the principal branch

$$\theta = \theta_{\rm raw} - \frac{\pi}{2}\,\mathrm{Round}\left(\frac{\theta_{\rm raw}}{\pi/2}\right), \qquad \theta \in [-45^\circ, 45^\circ]. \tag{18}$$

6 Momentum-space OPE

The perturbative heavy-quark propagator is

$$S_Q^{(0)}(k) = \frac{\not{k} + m_Q}{k^2 - m_Q^2}. \tag{19}$$

For the dimension-4 gluon condensate we use the background-field expansion

$$S_Q(k) = S_Q^{(0)}(k) + g_s G_A^{\alpha\beta} T^A S^{(G)}_{Q,\alpha\beta}(k) + S_Q^{(G^2)}(k), \tag{20}$$

where

$$S^{(G)}_{Q,\alpha\beta}(k) = -\frac{1}{4}\frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}, \tag{21}$$

$$S_Q^{(G^2)}(k) = \frac{\langle g_s^2 G^2\rangle}{12} m_Q \frac{k^2 + m_Q \not{k}}{(k^2 - m_Q^2)^4}. \tag{22}$$

Equivalently,

$$g_s G_A^{\alpha\beta} T^A S^{(G)}_{Q,\alpha\beta}(k) = -\frac{g_s G_A^{\alpha\beta} T^A}{4} \frac{\sigma_{\alpha\beta}(\not{k} + m_Q) + (\not{k} + m_Q)\sigma_{\alpha\beta}}{(k^2 - m_Q^2)^2}. \tag{23}$$

The one-gluon fields are contracted with

$$\langle g_s^2 G_A^{\alpha\beta} G_B^{\rho\sigma}\rangle = \frac{\delta_{AB}}{96}\langle g_s^2 G^2\rangle \left(g^{\alpha\rho}g^{\beta\sigma} - g^{\alpha\sigma}g^{\beta\rho}\right), \tag{24}$$

and

$$\mathrm{Tr}(T^A T^B) = \frac{\delta^{AB}}{2}. \tag{25}$$

Thus the color factor for the $S^{(G)}S^{(G)}$ piece is

$$\frac{N_c^2 - 1}{192}\langle g_s^2 G^2\rangle, \qquad N_c = 3. \tag{26}$$

The three projected OPE contributions are generated from

$$\Pi_1^{AA} \sim P_{(1)}^{\mu\nu}\,\mathrm{Tr}\left[\gamma_\mu\gamma_5 S_c(k)\gamma_\nu\gamma_5 S_b(k-p)\right], \tag{27}$$

$$\Pi_1^{AB} \sim P_{(1)}^{\mu\nu}\,\mathrm{Tr}\left[\gamma_\mu\gamma_5 S_c(k) i\sigma_{\nu\alpha}\frac{p^\alpha}{m_b+m_c}\gamma_5 S_b(k-p)\right], \tag{28}$$

$$\Pi_1^{BB} \sim P_{(1)}^{\mu\nu}\,\mathrm{Tr}\left[i\sigma_{\mu\alpha}\frac{p^\alpha}{m_b+m_c}\gamma_5 S_c(k) i\sigma_{\nu\beta}\frac{p^\beta}{m_b+m_c}\gamma_5 S_b(k-p)\right]. \tag{29}$$

7 Perturbative spectral densities

For the perturbative loop, impose the two-particle cut

$$k^2 = m_c^2, \qquad (k-p)^2 = m_b^2, \tag{30}$$

which gives

$$k\cdot p = \frac{s+m_c^2-m_b^2}{2}, \qquad s = p^2. \tag{31}$$

Define

$$\lambda(s,m_b,m_c) = s^2 + m_b^4 + m_c^4 - 2sm_b^2 - 2sm_c^2 - 2m_b^2m_c^2. \tag{32}$$

The perturbative spectral densities are implemented as

$$\rho_{\mathrm{pert}}^{ij}(s) = \frac{N_c}{16\pi^2}\frac{\sqrt{\lambda(s,m_b,m_c)}}{s}\frac{N^{ij}(s)}{(m_b+m_c)^{n_B^{ij}}}, \tag{33}$$

where $n_B^{AA}=0$, $n_B^{AB}=n_B^{BA}=1$, and $n_B^{BB}=2$ counts how many tensor currents occur in the channel. The unnormalized numerator functions are

$$N^{AA}(s) = -\frac{2}{s}\left[(m_b+m_c)^2 - s\right]\left[(m_b-m_c)^2+2s\right], \tag{34}$$

$$N^{AB}(s) = 6(m_b-m_c)\left[(m_b+m_c)^2-s\right], \tag{35}$$

$$N^{BB}(s) = 2\left[-2(m_b^2-m_c^2)^2 + (m_b^2-6m_bm_c+m_c^2)s + s^2\right]. \tag{36}$$

Equivalently, after setting $N_c=3$,

$$\rho_{\mathrm{pert}}^{AA}(s) = -\frac{3}{8\pi^2 s^2}\left[(m_b+m_c)^2-s\right]\left[(m_b-m_c)^2+2s\right]\sqrt{\lambda(s,m_b,m_c)}, \tag{37}$$

$$\rho_{\mathrm{pert}}^{AB}(s) = \frac{9}{8\pi^2 s(m_b+m_c)}(m_b-m_c)\left[(m_b+m_c)^2-s\right]\sqrt{\lambda(s,m_b,m_c)}, \tag{38}$$

$$\rho_{\mathrm{pert}}^{BB}(s) = \frac{3}{8\pi^2 s(m_b+m_c)^2}\left[-2(m_b^2-m_c^2)^2+(m_b^2-6m_bm_c+m_c^2)s+s^2\right]\sqrt{\lambda(s,m_b,m_c)}. \tag{39}$$

The perturbative Borel moments are

$$\Pi_{\mathrm{pert}}^{ij}(M^2,s_0) = \int_{(m_b+m_c)^2}^{s_0} ds\, e^{-s/M^2}\rho_{\mathrm{pert}}^{ij}(s). \tag{40}$$

With this convention all three perturbative spectral densities have mass dimension 2. Without the factors of $(m_b+m_c)^{-n_B^{ij}}$, their dimensions would be 2, 3, and 4 for AA, AB, and BB, respectively.

8 Dimension-4 and dimension-6 gluon condensates

The implemented condensate contribution is

$$\Pi^{ij}_{G^2} = \Pi^{ij}[S_c^{(G^2)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^2)}] + \Pi^{ij}[S_c^{(G)} S_b^{(G)}]. \tag{41}$$

After Feynman parametrization the terms have the schematic form

$$\mathcal{A}^{ij}_{G^2}(x_1, x_2, s) = \sum_r C_r^{ij}(x_1, x_2, s) \left[m_c^2 x_1^2 + (m_b^2 + m_c^2 - s)x_1 x_2 + m_b^2 x_2^2\right]^{-n_r}. \tag{42}$$

The code then fixes the simplex by

$$x_1 = x, \qquad x_2 = 1 - x. \tag{43}$$

This gives

$$\Delta(x, s) = m_c^2 x + m_b^2(1 - x) - s\, x(1 - x). \tag{44}$$

For the triple-gluon condensate we use the standard momentum-space vacuum-averaged heavy-quark propagator term

$$S_Q^{(G^3)}(k) = \frac{\langle g_s^3 G^3\rangle}{48} \frac{(\not{k} + m_Q)\left[\not{k}(k^2 - 3m_Q^2) + 2m_Q(2k^2 - m_Q^2)\right](\not{k} + m_Q)}{(k^2 - m_Q^2)^6}. \tag{45}$$

The implemented dimension-6 contribution is

$$\Pi^{ij}_{G^3,\mathrm{single}} = \Pi^{ij}[S_c^{(G^3)} S_b^{(0)}] + \Pi^{ij}[S_c^{(0)} S_b^{(G^3)}]. \tag{46}$$

This is the standard single-heavy-propagator contribution. The complete dimension–6 result would be

$$\Pi^{ij}_{G^3,\mathrm{complete}} = \Pi^{ij}_{G^3,\mathrm{single}} + \Pi^{ij}_{G^3,\mathrm{cross}}, \tag{47}$$

where

$$\Pi^{ij}_{G^3,\mathrm{cross}} = \Pi^{ij}[S_c^{(GG,\mathrm{open})} S_b^{(G,\mathrm{open})}] + \Pi^{ij}[S_c^{(G,\mathrm{open})} S_b^{(GG,\mathrm{open})}]. \tag{48}$$

The necessary vacuum average has the schematic form

$$\left\langle g_s^3 G_A^{\alpha\beta} G_B^{\rho\sigma} G_C^{\lambda\tau}\right\rangle = \frac{f_{ABC}}{576} \left\langle g_s^3 G^3\right\rangle T^{\alpha\beta,\rho\sigma,\lambda\tau}, \tag{49}$$

$$\begin{aligned} T^{\alpha\beta,\rho\sigma,\lambda\tau} &= g^{\alpha\rho}(g^{\beta\lambda}g^{\sigma\tau} - g^{\beta\tau}g^{\sigma\lambda}) - g^{\alpha\sigma}(g^{\beta\lambda}g^{\rho\tau} - g^{\beta\tau}g^{\rho\lambda}) \\ &\quad - g^{\beta\rho}(g^{\alpha\lambda}g^{\sigma\tau} - g^{\alpha\tau}g^{\sigma\lambda}) + g^{\beta\sigma}(g^{\alpha\lambda}g^{\rho\tau} - g^{\alpha\tau}g^{\rho\lambda}). \end{aligned} \tag{50}$$

The coefficient is fixed by the candidate normalization

$$f^{ABC}\left\langle g_s^3 G_A^{\mu\nu} G_B^{\nu\rho} G_C^{\rho\mu}\right\rangle = \left\langle g_s^3 G^3\right\rangle, \qquad T_{\mu\nu,\nu\rho,\rho\mu} = 24, \quad f^{ABC}f^{ABC} = 24 \quad (N_c = 3). \tag{51}$$

Before using this numerically, the sign and index order must be checked against the convention used for $\langle g_s^3 G^3\rangle$. Most importantly, $S_Q^{(GG,\mathrm{open})}$ is not obtained by "undoing" the averaged $S_Q^{(G^2)}$ propagator; it must be derived or imported as an explicitly open-field two-gluon propagator. This is why the present numerical notebooks quote G^3_{single}, not G^3_{complete}.

The active implementation path is now encoded in `BcMixingDimension6Complete.wl`. The required fixed-point-gauge derivation is:

1. Use $x^\mu A_\mu(x) = 0$ and expand $A_\mu^A(y) = \frac{1}{2} y^\alpha G^A_{\alpha\mu}(0) + \cdots$.

2. In momentum space replace y^α by $i\partial/\partial k_\alpha$.

3. Expand the heavy propagator recursively, $S = S_0 + S_1 + S_2 + \cdots$, keeping the color order in S_2 as $T^A T^B$ and $T^B T^A$.

4. Fix all signs and factors by requiring S_1 to reproduce the one-gluon propagator used in the momentum-space code.

5. Vacuum-average the two gluons inside S_2. This same-line contraction must reproduce the implemented
$$S_Q^{(G^2)}(k) = \frac{\langle g_s^2 G^2 \rangle}{12} m_Q \frac{k^2 + m_Q \not{k}}{(k^2 - m_Q^2)^4}.$$

6. Only after this calibration may the cross-line products $S_c^{(GG,\text{open})} S_b^{(G,\text{open})}$ and $S_c^{(G,\text{open})} S_b^{(GG,\text{open})}$ be reduced to Borel moments.

This protects the calculation from a common error: using an averaged G^2 propagator as if it still contained the color and Lorentz information needed for a triple-gluon cross-line contraction.

The Wolfram file now contains an experimental implementation of this derivation route:

```
Get["BcMixingMomentum.wl"];
Get["BcMixingDimension6Complete.wl"];
D6OneGluonCalibrationResidual[k, mc, al, be, Cins]
D6OpenTwoGluonCandidate[k, mc, al, be, rh, si, Cins]
```

The first line is the decisive test. A candidate insertion operator is not accepted until the residual vanishes after fixing the insertion coefficient `Cins` and simplifying in the same Dirac/Feynman-denominator convention used by the momentum-space calculation. The present simple fixed-point-gauge repeated-insertion ansatz does not yet pass this test; hence no paper-final G^3_{cross} moment is quoted in the production notebooks. This is a calculation result, not merely a missing switch: the color-antisymmetric part needed by $f^{ABC}\langle G_A G_B G_C \rangle$ is not fixed by the same-line G^2 propagator.

As an exploratory check, the workbench now reduces one side of the cross-line operator through the full sequence
$$S_c^{(GG,\text{open})} S_b^{(G,\text{open})} \to P_1^{\mu\nu} \Pi_{\mu\nu} \to \mathcal{A}(x, s) \to \widehat{\mathcal{B}}_{M^2} \mathcal{A}.$$

For the AA channel at the central Wang-window point,
$$M^2 = 8.0 \text{ GeV}^2, \qquad s_0 = 54.0 \text{ GeV}^2,$$

the tested side gives
$$\Pi^{AA}_{G^3,\text{cross};\, c2b1} = -1.6756975780964764 \times 10^{-6}. \tag{52}$$

The mirror side is more expensive, but it can be evaluated with cached termwise traces and Feynman-parameter chunks. The corresponding central value is
$$\Pi^{AA}_{G^3,\text{cross};\, c1b2} = 3.378713957299235 \times 10^{-7}. \tag{53}$$

Thus the currently tested AA cross-line sum is
$$\Pi^{AA}_{G^3,\text{cross}} = -1.337826182366553 \times 10^{-6} \qquad (M^2 = 8.0 \text{ GeV}^2,\ s_0 = 54.0 \text{ GeV}^2). \tag{54}$$

The AB and BB reductions have also been completed with the same cached termwise method. At the same central point the cross-line moments are

ij	$\Pi^{ij}_{G^3,\text{cross};\, c2b1}$	$\Pi^{ij}_{G^3,\text{cross};\, c1b2}$	$\Pi^{ij}_{G^3,\text{cross}}$
AA	$-1.6756975780964764 \times 10^{-6}$	$3.378713957299235 \times 10^{-7}$	$-1.337826182366553 \times 10^{-6}$
AB	$6.907526226109287 \times 10^{-7}$	$1.192136959955478 \times 10^{-7}$	$8.099663186064834 \times 10^{-7}$
BB	$-1.168414043165516 \times 10^{-6}$	$-7.557182424790687 \times 10^{-7}$	$-1.924132285644585 \times 10^{-6}$

Adding these cached cross-line moments to the current production moments gives

$$\theta_{\text{single-line }G^3} = 43.310272930706944^\circ, \qquad \theta_{\text{with cross-line}} = 43.309988657908924^\circ,$$

so that

$$\Delta\theta_{G^3,\text{cross}} = -2.8427279801945815 \times 10^{-4}\ \text{deg}. \tag{55}$$

These numbers should still be read as validation of the algebraic workbench: the one-gluon calibration convention still needs a final audit. Therefore the current paper-facing "total" result remains

$$\Pi_{\text{total}} = \Pi_{\text{pert}} + \Pi_{G^2} + \Pi_{G^3,\text{single}},$$

not $\Pi_{G^3,\text{complete}}$. The current Mathematica status can be queried with `D6CrossLineValidationStatus[]`.

Operationally, the momentum-space notebook `BcMixingMomentum.nb` is the file to run for the production momentum results. In that notebook the order `"total"` means

$$\Pi_{\text{pert}} + \Pi_{G^2} + \Pi_{G^3,\text{single}}.$$

The switch `showCachedCompleteD6CentralQuote` only displays or quotes the cached central cross-line correction at $M^2 = 8\ \text{GeV}^2$, $s_0 = 54\ \text{GeV}^2$. It is not a substitute for a full M^2- and s_0-dependent cross-line grid. Similarly, the Monte Carlo switch `requestCompleteD6CrossLineMonteCarlo` is kept `False` by default; a point-by-point uncertainty propagation of $\Pi_{G^3,\text{cross}}$ requires generating a cross-line grid or interpolation first.

For the paper-style $\theta(M^2)$ decomposition plot with the complete dimension–6 correction, the notebook contains a separate `Complete-D6 M2 Grid` block. One first sets

```
completeD6PlotS0 = 53.0;
completeD6GridNPoints = stabilityNPoints;
generateCompleteD6CrossLineM2Grid = True;
overwriteCompleteD6CrossLineM2Grid = False;
```

and evaluates the grid cell. The calculation stores each point in `d6_chunks`; an interrupted calculation can therefore be resumed by rerunning the same cell. Once the grid is available, set

```
generateCompleteD6CrossLineM2Grid = False;
```

and the notebook loads the cache and plots four curves:

$$\theta_{\text{pert}}, \qquad \theta_{\text{pert+G}^2}, \qquad \theta_{\text{pert+G}^2+\text{G}^3_{\text{single}}}, \qquad \theta_{\text{pert+G}^2+\text{G}^3_{\text{single}}+\text{G}^3_{\text{cross}}}.$$

The product

$$S_c^{(GG,\text{open})} S_b^{(GG,\text{open})}$$

is not part of the dimension–6 G^3 basis. It contains four background field strengths and therefore belongs to the dimension–8 OPE sector, schematically $\langle G^4\rangle$, or under factorization $\langle G^2\rangle^2$. It should be considered only in a higher-dimensional OPE extension, not in G^3_{complete}.

The G^2 and G^3 terms are not treated as ordinary smooth spectral densities. We instead use a direct Borel transform. Set $s = -Q^2$ and write

$$\Delta(x, -Q^2) = x(1-x)\left[Q^2 + \bar{s}(x)\right], \tag{56}$$

where

$$\bar{s}(x) = \frac{m_c^2 x + m_b^2(1-x)}{x(1-x)}. \tag{57}$$

The Feynman-parameter amplitude is decomposed as

$$-i\,\mathcal{A}^{ij}_{G^r}(x, -Q^2) = \sum_{\ell\geq 1} \frac{a^{ij}_\ell(x)}{[Q^2 + \bar{s}(x)]^\ell} + \text{polynomial subtraction terms}. \tag{58}$$

The polynomial subtraction terms vanish after the Borel transform. The working phase convention is the explicit factor $-i$ shown above. This is a bookkeeping convention for removing the loop-integration prefactor returned by FeynCalc's Feynman-parameter amplitude. This is the convention that should be independently checked.

We use

$$\mathcal{B}_{M^2}\left[\frac{1}{(Q^2+\bar{s})^n}\right] = \frac{1}{(n-1)!\,(M^2)^{n-1}} e^{-\bar{s}/M^2}. \tag{59}$$

Thus

$$\Pi^{ij}_{G^r}(M^2, s_0) = \int_{x_-}^{x_+} dx\, e^{-\bar{s}(x)/M^2} \sum_{\ell\geq 1} \frac{a^{ij}_\ell(x)}{(\ell-1)!\,(M^2)^{\ell-1}}. \tag{60}$$

The continuum restriction is

$$\bar{s}(x) \leq s_0, \tag{61}$$

which gives

$$x_\pm = \frac{s_0 + m_b^2 - m_c^2 \pm \sqrt{\lambda(s_0, m_b, m_c)}}{2s_0}. \tag{62}$$

9 Total Borel moments and numerical test point

The total moments used in the current numerical result are

$$\Pi^{ij}(M^2, s_0) = \Pi^{ij}_{\text{pert}}(M^2, s_0) + \Pi^{ij}_{G^2}(M^2, s_0) + \Pi^{ij}_{G^3}(M^2, s_0). \tag{63}$$

The current default numerical inputs are

$$m_b = 4.18 \text{ GeV}, \tag{64}$$
$$m_c = 1.27 \text{ GeV}, \tag{65}$$
$$\langle g_s^2 G^2\rangle = 4\pi^2(0.012) \text{ GeV}^4, \tag{66}$$
$$\langle g_s^3 G^3\rangle = 0.57 \text{ GeV}^6, \tag{67}$$
$$M^2 = 10 \text{ GeV}^2, \tag{68}$$
$$s_0 = 55 \text{ GeV}^2. \tag{69}$$

For this point,

$$x_- = 0.3322728664, \qquad x_+ = 0.9560816791. \tag{70}$$

The perturbative moments are

$$\Pi^{AA}_{\text{pert}} = 0.1441716826, \tag{71}$$
$$\Pi^{AB}_{\text{pert}} = -0.1051577192, \tag{72}$$
$$\Pi^{BB}_{\text{pert}} = 0.1348134513. \tag{73}$$

The gluon-condensate moments are

$$\Pi^{AA}_{G^2} = -0.0003658277, \tag{74}$$
$$\Pi^{AB}_{G^2} = 0.0002309420, \tag{75}$$
$$\Pi^{BB}_{G^2} = -0.0001888869. \tag{76}$$

The triple-gluon-condensate moments are

$$\Pi^{AA}_{G^3} = 0.0000700979, \tag{77}$$
$$\Pi^{AB}_{G^3} = -0.0000493659, \tag{78}$$
$$\Pi^{BB}_{G^3} = 0.0000310238. \tag{79}$$

Thus

$$\Pi^{AA}_{\text{total}} = 0.1438759528, \tag{80}$$
$$\Pi^{AB}_{\text{total}} = -0.1049761432, \tag{81}$$
$$\Pi^{BB}_{\text{total}} = 0.1346555882. \tag{82}$$

The corresponding angles are

$$\theta_{\text{pert}} = 43.726118943^\circ, \tag{83}$$
$$\theta_{\text{pert+G}^2} = 43.747426660^\circ, \tag{84}$$
$$\theta_{\text{pert+G}^2\text{+G}^3} = 43.742693524^\circ. \tag{85}$$

At this point the relative condensate corrections are

$$\frac{\Pi^{AA}_{G^2}}{\Pi^{AA}_{\text{pert}}} = -0.002537, \tag{86}$$
$$\frac{\Pi^{AB}_{G^2}}{\Pi^{AB}_{\text{pert}}} = -0.002196, \tag{87}$$
$$\frac{\Pi^{BB}_{G^2}}{\Pi^{BB}_{\text{pert}}} = -0.001401, \tag{88}$$
$$\frac{\Pi^{AA}_{G^3}}{\Pi^{AA}_{\text{pert}}} = 0.000486, \tag{89}$$
$$\frac{\Pi^{AB}_{G^3}}{\Pi^{AB}_{\text{pert}}} = 0.000469, \tag{90}$$
$$\frac{\Pi^{BB}_{G^3}}{\Pi^{BB}_{\text{pert}}} = 0.000230. \tag{91}$$

10 Perturbative α_s sensitivity model

The independent $O(\alpha_s)$ correction to the AA channel has now been calculated in `BcMixingAlphaS.wl`. The corresponding AB and BB tensor-current corrections are not yet available. Before the explicit calculation was started, the code used the following channel-dependent trial rescaling of the leading perturbative moments:

$$\Pi^{ij}_{\text{pert}}(M^2, s_0) \longrightarrow \left[1 + \frac{\alpha_s}{\pi} K_{ij}\right] \Pi^{ij}_{\text{pert}}(M^2, s_0). \tag{92}$$

The condensate pieces are left unchanged, so for the current total moment

$$\Pi^{ij}_{K} = \left[1 + \frac{\alpha_s}{\pi} K_{ij}\right] \Pi^{ij}_{\text{pert}} + \Pi^{ij}_{G^2} + \Pi^{ij}_{G^3}. \tag{93}$$

This remains useful as a *sensitivity model*, but it is not a calculation of a two-loop spectral density. A master's student should distinguish the following questions:

1. The explicit NLO calculation asks: what functions $\rho_1^{AA}(s)$, $\rho_1^{AB}(s)$, and $\rho_1^{BB}(s)$ follow from QCD?

2. The K_{ij} model asks: how much would the mixing angle move if the three moments were rescaled by selected constants?

The model cannot check the threshold dependence, logarithms, mass scheme, real–virtual cancellation, or continuum-subtracted shape of the true NLO spectral density. Its value is that it exposes cancellations in

$$\tan(2\theta) = \frac{-2\Pi^{AB}}{\Pi^{AA} - \Pi^{BB}}. \tag{94}$$

If $K_{AA} = K_{AB} = K_{BB}$, the perturbative pieces receive nearly a common factor and much of the correction cancels from the angle. Large shifts arise when the channel corrections are different. Therefore the scan estimates the possible *angle sensitivity*; it does not estimate the size or correctness of the AA correction.

For example, with $\alpha_s = 0.26$ and

$$K_{AA} = 1, \qquad K_{AB} = 0, \qquad K_{BB} = -1, \tag{95}$$

the total angle at $M^2 = 10$ GeV^2, $s_0 = 55$ GeV^2 changes from

$$\theta_{\text{base}} = 43.742693524^\circ \tag{96}$$

to

$$\theta_K = 40.625721673^\circ, \qquad \Delta\theta = -3.116971851^\circ. \tag{97}$$

A deliberately broad coarse scan with $K_{AA}, K_{AB}, K_{BB} \in \{-2, 0, 2\}$ gives a maximum branch-corrected shift of about

$$\max|\Delta\theta| \simeq 7.52^\circ. \tag{98}$$

This result is helpful because it shows that channel-to-channel differences can move the angle by several degrees. It should not be quoted as a QCD uncertainty distribution, because the three K_{ij} values are neither derived nor statistically distributed.

11 Monte Carlo uncertainty propagation

The numerical input table gives central values,

$$m_b = 4.18 \text{ GeV}, \qquad m_c = 1.27 \text{ GeV}, \tag{99}$$

$$\langle g_s^2 G^2 \rangle = 4\pi^2(0.012) \text{ GeV}^4, \qquad \langle g_s^3 G^3 \rangle = 0.57 \text{ GeV}^6. \tag{100}$$

For the final quoted error these inputs should be assigned allowed intervals or probability distributions. In the present implementation we use uniform sampling over editable intervals. A Monte Carlo point is

$$X_i = \left\{ m_b^{(i)}, m_c^{(i)}, G_2^{(i)}, G_3^{(i)}, M_i^2, s_{0,i} \right\}, \tag{101}$$

where

$$G_2 \equiv \langle g_s^2 G^2 \rangle, \qquad G_3 \equiv \langle g_s^3 G^3 \rangle. \tag{102}$$

Each component is drawn from the user-specified interval

$$x^{(i)} \sim U(x_{\min}, x_{\max}). \tag{103}$$

Fixed parameters are implemented by setting $x_{\min} = x_{\max}$, or in the Mathematica code by passing a single number instead of an interval.

The current working ranges in the code are

$$m_b \in [4.16, 4.21] \text{ GeV}, \qquad m_c \in [1.25, 1.29] \text{ GeV}, \tag{104}$$

$$G_2 \in 4\pi^2[0.006, 0.018] \text{ GeV}^4, \qquad G_3 \in [0.28, 0.86] \text{ GeV}^6, \tag{105}$$

$$M^2 \in [8, 12] \text{ GeV}^2, \qquad s_0 \in [50, 60] \text{ GeV}^2. \tag{106}$$

Equivalently, in the central-value-plus-error notation useful for the paper,

$$m_b = (4.185 \pm 0.025)\ \text{GeV}, \qquad m_c = (1.27 \pm 0.02)\ \text{GeV}, \tag{107}$$

$$G_2 = 4\pi^2(0.012 \pm 0.006)\ \text{GeV}^4, \qquad G_3 = (0.57 \pm 0.29)\ \text{GeV}^6, \tag{108}$$

$$M^2 = (10 \pm 2)\ \text{GeV}^2, \qquad s_0 = (55 \pm 5)\ \text{GeV}^2. \tag{109}$$

These are not final physics choices. They are placeholders for the scan layer and should be replaced by the intervals adopted in the analysis.

For every random point we impose the physical continuum condition

$$s_{0,i} > \left(m_b^{(i)} + m_c^{(i)}\right)^2. \tag{110}$$

Accepted points are evaluated with the same OPE formula as above,

$$\theta_i = \frac{1}{2}\,\text{ArcTan}\left[\Pi_1^{AA}(X_i) - \Pi_1^{BB}(X_i), -2\Pi_1^{AB}(X_i)\right], \tag{111}$$

using the chosen OPE order. The paper-facing momentum-space uncertainty scan now uses "total",

$$\Pi_{\text{MC}}^{ij} = \Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij} + \Pi_{G^3,\text{single}}^{ij},$$

by setting `"IncludeG3" -> True` in the Monte Carlo call. The package-level default

```
$BcMixingMonteCarloIncludeG3 = False
```

remains conservative so that quick exploratory scans do not accidentally evaluate the slower G^3 Borel moments. With `"IncludeG3" -> False`, a requested "total" Monte Carlo scan is evaluated as "pertG2",

$$\Pi_{\text{MC}}^{ij} = \Pi_{\text{pert}}^{ij} + \Pi_{G^2}^{ij}, \tag{112}$$

which is useful only as a diagnostic comparison.

The central value and Gaussian-width estimate are then

$$\bar{\theta} = \frac{1}{N}\sum_{i=1}^{N}\theta_i, \qquad \sigma_{\theta,\text{Gauss}} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(\theta_i - \bar{\theta}\right)^2}. \tag{113}$$

The code also reports the unbiased sample standard deviation, median, minimum, maximum, and the 16% and 84% quantiles. These quantiles are useful because the resulting θ distribution may not be exactly Gaussian.

The Mathematica entry point is

```
nSamples = 500;
histBins = 25;
m2Range = {8.0, 12.0};
s0Range = {50.0, 60.0};

ranges = Join[
  $BcMixingDefaultUncertaintyRanges,
  <|"M2" -> m2Range, "s0" -> s0Range|>
];

uncertaintyRun = MonteCarloMixingAngleUncertainty[
  nSamples, ranges, "total",
  "IncludeG3" -> False,
```

```
  "Seed" -> 1234,
  "Progress" -> True
];
uncertaintyRun["Summary"]
MonteCarloMixingAngleHistogram[uncertaintyRun, histBins]
MonteCarloMixingAnglePublicationHistogram[uncertaintyRun, histBins]
```

where `ranges` is an association such as

```
<|
  "mb" -> {4.16, 4.21},
  "mc" -> {1.25, 1.29},
  "G2" -> {4 Pi^2 0.006, 4 Pi^2 0.018},
  "G3" -> {0.28, 0.86},
  "M2" -> {8.0, 12.0},
  "s0" -> {50.0, 60.0}
|>
```

The number of random points, the input intervals, and the Borel window are therefore all editable without touching the algebra.

The first argument, for example `500`, is the number of random accepted Monte Carlo points. The option `"Seed" -> 1234` fixes the random-number seed so the sample is reproducible. The Monte Carlo result is called `uncertaintyRun`, not `mc`, to avoid confusion with the charm mass parameter `"mc"`. To vary only the Borel window and the continuum threshold while keeping the QCD inputs fixed at their central values, one can use

```
ranges = Join[
  $BcMixingDefaultUncertaintyRanges,
  <|
    "mb" -> 4.18,
    "mc" -> 1.27,
    "G2" -> 4 Pi^2 0.012,
    "G3" -> 0.57,
    "M2" -> {8.0, 12.0},
    "s0" -> {50.0, 60.0}
  |>
];
```

The accepted random points and the summary can be exported for replotting:

```
ExportMonteCarloMixingAngleSamples[uncertaintyRun, "BcMixingMonteCarloSamples.csv"];
ExportMonteCarloMixingAngleSummary[uncertaintyRun, "BcMixingMonteCarloSummary.csv"];
Export["BcMixingMonteCarloHistogram.pdf",
  MonteCarloMixingAnglePublicationHistogram[uncertaintyRun, 25]];
```

12 How to reproduce the numerical check

In Mathematica, load the implementation with

```
SetDirectory["/Users/sbilmis/Bc_mixing"];
Get["BcMixingMomentum.wl"];
```

Then evaluate

```
NumericOPESummary[10, 55]
NumericMixingAngleDegrees[10, 55, "pert"]
NumericMixingAngleDegrees[10, 55, "pertG2"]
NumericMixingAngleDegrees[10, 55, "total"]
KFactorSensitivityRecord[10, 55, KFactorAssociation[1, 0, -1], 0.26, "total"]
KFactorSensitivityEnvelope[10, 55, {-2, 2, 2}, 0.26, "total"]
MixingMassDimensionReport[]
CheckMixingMassDimensions[]
PerturbativeSpectralDensityDimensionReport[]
```

For window studies one can scan

```
MixingAngleTable[{8, 12, 1}, {50, 60, 5}, "total"]
OPEConvergenceDataset[{8, 12, 1}, {50, 60, 5}]
uncertaintyRun = MonteCarloMixingAngleUncertainty[50, $BcMixingDefaultUncertaintyRanges,
  "total", "IncludeG3" -> False, "Seed" -> 1234]
uncertaintyRun["Summary"]
```

For paper presentation the most meaningful decomposition plot is the physical mixing angle itself. We therefore plot $\theta(M^2)$ at fixed s_0 for the perturbative, perturbative plus G^2, and total OPE orders:

```
thetaOrderM2 = MixingAngleOrderM2PublicationPlot[
  $BcMixingWangWindow["M2Range"],
  53.0,
  {"pert", "pertG2", "total"},
  $BcMixingDefaultParameters,
  "NPoints" -> 25,
  "YHalfWidth" -> 1.0
];

thetaOrderM2["Plot"]
thetaOrderM2["Data"] // Dataset
```

The channel-level OPE hierarchy is then shown separately through

$$\left|\frac{\Pi^{ij}_{G^2}}{\Pi^{ij}_{\rm pert}}\right|, \qquad \left|\frac{\Pi^{ij}_{G^3}}{\Pi^{ij}_{\rm pert}}\right|, \qquad ij = AA, AB, BB.$$

This separation is important: AA, AB, and BB are not separate mixing-angle predictions; they are the three invariant amplitudes entering the single mixing-angle formula. The diagnostic plot is generated by

```
momentumRatioM2 = OPEContributionRatioM2PublicationPlot[
  $BcMixingWangWindow["M2Range"],
  53.0,
  $BcMixingDefaultParameters,
  "NPoints" -> 25,
  "UseAbs" -> True
];

momentumRatioM2["Plot"]
momentumRatioM2["Data"] // Dataset
```

Setting `"UseAbs" -> False` displays the signed ratios

$$\frac{\Pi^{ij}_{G^2}}{\Pi^{ij}_{\rm pert}}, \qquad \frac{\Pi^{ij}_{G^3}}{\Pi^{ij}_{\rm pert}}. \tag{114}$$

13 Auxiliary-parameter stability plots

As a comparison with the vector B_c^* analysis of Wang, we also consider the auxiliary window

$$M^2 \in [7.0, 9.0]\ \mathrm{GeV}^2, \qquad s_0 = (54 \pm 1)\ \mathrm{GeV}^2. \tag{115}$$

The Mathematica implementation stores this as

```
$BcMixingWangWindow
```

and produces publication-style stability plots with

```
stabilityOrder = "total";

m2Stability = MixingAngleStabilityM2PublicationPlot[
  $BcMixingWangWindow["M2Range"],
  $BcMixingWangWindow["s0Values"],
  stabilityOrder,
  $BcMixingDefaultParameters,
  "NPoints" -> 25,
  "YHalfWidth" -> 1.0
];

s0Stability = MixingAngleStabilityS0PublicationPlot[
  $BcMixingWangWindow["s0Range"],
  $BcMixingWangWindow["M2Values"],
  stabilityOrder,
  $BcMixingDefaultParameters,
  "NPoints" -> 25,
  "YHalfWidth" -> 1.0
];
```

The option `"YHalfWidth" -> 1.0` prevents the vertical axis from being over-zoomed; it enforces at least a $\pm 1^\circ$ display window around the calculated angles. This is useful when the goal is to demonstrate stability with respect to the auxiliary parameters, not to magnify tiny numerical variations. For publication labels the plotting functions try to use the Mathematica package MaTeX automatically,

```
"UseMaTeX" -> Automatic
```

so that the axes and legends are rendered as $M^2(\mathrm{GeV}^2)$, $s_0(\mathrm{GeV}^2)$, θ°, and entries such as $s_0 = 54\,\mathrm{GeV}^2$. If MaTeX is not installed, the code falls back to ordinary Mathematica labels with a degree symbol. The fallback can be forced with `"UseMaTeX" -> False`.

The plots and their underlying data can be exported with

```
Export["BcMixingThetaVsM2_WangWindow.pdf", m2Stability["Plot"]];
Export["BcMixingThetaVsS0_WangWindow.pdf", s0Stability["Plot"]];
ExportStabilityDataCSV[m2Stability["Data"],
  "BcMixingThetaVsM2_WangWindow.csv", "M2"];
ExportStabilityDataCSV[s0Stability["Data"],
  "BcMixingThetaVsS0_WangWindow.csv", "s0"];
```

14 Independent coordinate-space route

The file `BcMixingCoordinate.wl` defines an independent coordinate-space route. In the current Mathematica workflow the notebook `BcMixingCoordinate.nb` loads only this coordinate file in its main calculation path. Momentum-space comparisons are kept in an optional commented section and should not be used to generate coordinate-space paper-final numbers.

At present, the independent coordinate-space result is the perturbative spectral density and its Borel moments. The same interpolating currents and the same normalization of the tensor current are used,

$$J_\mu^A(x) = \bar{b}(x)\gamma_\mu\gamma_5 c(x), \qquad J_\mu^B(x) = i\bar{b}(x)\sigma_{\mu\alpha}\frac{p^\alpha}{m_b+m_c}\gamma_5 c(x). \tag{116}$$

With these conventions, the coordinate-space perturbative correlators are

$$\Pi_{\mu\nu}^{AA}(p) = -iN_c\int d^4x\, e^{ip\cdot x}\mathrm{Tr}\left[\gamma_\mu\gamma_5 S_c^{(0)}(-x)\gamma_\nu\gamma_5 S_b^{(0)}(x)\right], \tag{117}$$

$$\Pi_{\mu\nu}^{AB}(p) = -iN_c\int d^4x\, e^{ip\cdot x}\mathrm{Tr}\left[\gamma_\mu\gamma_5 S_c^{(0)}(-x)i\sigma_{\nu\beta}\frac{p^\beta}{m_b+m_c}\gamma_5 S_b^{(0)}(x)\right], \tag{118}$$

$$\Pi_{\mu\nu}^{BB}(p) = -iN_c\int d^4x\, e^{ip\cdot x}\mathrm{Tr}\left[i\sigma_{\mu\alpha}\frac{p^\alpha}{m_b+m_c}\gamma_5 S_c^{(0)}(-x)i\sigma_{\nu\beta}\frac{p^\beta}{m_b+m_c}\gamma_5 S_b^{(0)}(x)\right]. \tag{119}$$

The sign difference between $S_c(-x)$ and $S_b(x)$ is implemented in the code by changing the sign of the $\not{x}$ term. The spin–1 invariant amplitude is extracted with the same transverse projector as in momentum space,

$$\Pi_1^{ij}(p^2) = \frac{1}{3}\left(g^{\mu\nu} - \frac{p^\mu p^\nu}{p^2}\right)\Pi_{\mu\nu}^{ij}(p). \tag{120}$$

This step is performed in `CoordinateTraceKernel[channel,"pert"]`.

The free heavy-quark propagator is written in terms of modified Bessel functions,

$$S_Q^{(0)}(x) = \frac{m_Q^2}{(2\pi)^2}\left[i\not{x}\frac{K_2(m_Q\sqrt{-x^2})}{-x^2} + \frac{K_1(m_Q\sqrt{-x^2})}{\sqrt{-x^2}}\right]. \tag{121}$$

This is the form used in the coordinate-space treatments of heavy systems. The derivative-free perturbative trace contains products of the form

$$(\not{x})^a(p\cdot x)^b\frac{K_\nu(m_c\sqrt{-x^2})K_\mu(m_b\sqrt{-x^2})}{(\sqrt{-x^2})^n}, \tag{122}$$

where the tensor numerator is reduced by FeynCalc before the radial Fourier integral is performed.

After Wick rotation to Euclidean momentum $Q^2 = -p^2$, the angular integral has the standard form

$$i\int d^Dx\, e^{ip\cdot x}f(\sqrt{-x^2}) = (2\pi)^{D/2}Q^{1-D/2}\int_0^\infty dx\, x^{D/2}J_{D/2-1}(Qx)f(x). \tag{123}$$

Therefore every scalar term is reduced to radial integrals

$$I_{u\nu\mu}(Q;m_c,m_b) = \int_0^\infty dx\, x^{u-1}J_v(Qx)K_\nu(m_c x)K_\mu(m_b x). \tag{124}$$

For equal heavy masses, Huang–Liu give a compact hypergeometric expression for this integral. For two different heavy masses, the Azizi et al. Bessel/Schwinger representation leads to the same physical two-body cut. The installed perturbative spectral density is

$$\rho_{\mathrm{pert}}^{ij}(s) = \frac{N_c}{16\pi^2}\frac{\sqrt{\lambda(s,m_b,m_c)}}{s}\,N^{ij}(s)\,\left(m_b+m_c\right)^{-n_{ij}}, \tag{125}$$

where $n_{AA} = 0$, $n_{AB} = 1$, $n_{BB} = 2$, and

$$\lambda(s, m_b, m_c) = s^2 + m_b^4 + m_c^4 - 2sm_b^2 - 2sm_c^2 - 2m_b^2 m_c^2, \tag{126}$$

$$k \cdot p\big|_{\text{cut}} = \frac{s + m_c^2 - m_b^2}{2}. \tag{127}$$

The numerator functions are

$$N^{AA}(s) = -\frac{4}{s}\left[2(k \cdot p)^2 + (3m_b m_c + m_c^2)s - 3s(k \cdot p)\right], \tag{128}$$

$$N^{AB}(s) = 12\left[-(m_b + m_c)(k \cdot p) + m_c s\right], \tag{129}$$

$$N^{BB}(s) = -4\left[4(k \cdot p)^2 + (3m_b m_c - m_c^2)s - 3s(k \cdot p)\right]. \tag{130}$$

A hand check should verify these three numerators from the projected coordinate-space trace and then verify the phase-space factor $N_c\sqrt{\lambda}/(16\pi^2 s)$. In the code these expressions are stored in `CoordinatePerturbativeNumerator` and `CoordinatePerturbativeSpectralDensity`.

The perturbative Borel moment is then evaluated as

$$\Pi^{ij}_{\text{pert}}(M^2, s_0) = \int_{(m_b+m_c)^2}^{s_0} ds\, e^{-s/M^2} \rho^{ij}_{\text{pert}}(s), \tag{131}$$

and the angle is obtained from the same diagonalization formula,

$$\theta^{\text{pert}}_{\text{coord}}(M^2, s_0) = \frac{1}{2}\,\text{ArcTan}\left[\Pi^{AA}_{\text{pert}} - \Pi^{BB}_{\text{pert}}, -2\Pi^{AB}_{\text{pert}}\right]. \tag{132}$$

14.1 What should be written in the journal paper

If the paper presents the coordinate-space calculation as the main method, the main text should not merely state that the result was obtained with Mathematica. It should show enough of the analytic reduction that a reader can see where the physical cut and continuum subtraction enter. A compact but sufficient presentation would include:

1. the two currents, including the tensor-current normalization $p^\alpha/(m_b + m_c)$;
2. the coordinate-space correlators after Wick contraction;
3. the heavy-quark propagator in terms of $K_\nu(m_Q\sqrt{-x^2})$;
4. the spin–1 projection used to define Π^{ij}_1;
5. the generic Bessel integral generated by the Fourier transform;
6. the way the Bessel functions are converted to Schwinger-parameter integrals;
7. the emergence of Dirac delta functions after the Borel transform;
8. the resulting Heaviside support conditions or, equivalently, the integration limits that implement continuum subtraction;
9. the final perturbative spectral densities, preferably in the compact $N^{ij}(s)\sqrt{\lambda}/s$ form;
10. the statement that the current coordinate-space condensate check includes local G^2 and G^3 single-line terms, while the full dimension–4 coordinate-space result additionally needs the $S_c^{(G)} S_b^{(G)}$ cross-line term.

The main text can contain the compact formulas and the final spectral densities. Longer channel-by-channel trace expressions and the local condensate Borel kernels are better placed in an appendix or supplemental file.

14.2 Role of the Azizi et al. method

The main technical role of the Azizi et al. method is to make continuum subtraction well defined for systems with more than one heavy propagator in coordinate space. With one heavy quark, the coordinate-space Fourier transform contains one modified Bessel function and can often be reduced to ordinary hypergeometric functions. With two heavy quarks, the Fourier transform contains products

$$J_v(Qx)K_\nu(m_c x)K_\mu(m_b x), \tag{133}$$

and a naive treatment of the Bessel functions can lead to parameter integrals with apparent endpoint divergences or ambiguous continuum subtraction.

Azizi et al. avoid this by using an integral representation of the modified Bessel functions and then applying the Borel transformations before doing all parameter integrations. In this order of operations, the Borel transforms generate Dirac delta functions. These delta functions fix combinations of Schwinger parameters and convert the would-be divergent parameter domains into finite domains with explicit support conditions. The support conditions are usually written as Heaviside functions or, after solving the inequalities, as modified integration limits.

Without this organization one would have to handle the delta functions, derivatives of delta functions, endpoint constraints, and continuum subtraction analytically by hand for each term. That is possible, but it is error-prone for two heavy masses because every numerator power changes the derivative order of the delta function or the power of the support factor. The Azizi formulation gives a systematic recipe:

$$\begin{aligned} K_\nu K_\mu &\longrightarrow \text{Schwinger parameters} \longrightarrow \text{Borel delta functions} \\ &\longrightarrow \Theta\text{-functions} \longrightarrow \text{finite integration limits.} \end{aligned} \tag{134}$$

In the paper, this chain should be shown once for a generic term. The final channel-dependent spectral densities can then be quoted in compact form.

14.3 Direct coordinate-space route without the Azizi reduction

It is useful to state explicitly what would happen if one deliberately did not use the Azizi et al. organization. This is not the route used by the current numerical code, and it should be treated as an independent analytic cross-check or appendix-level exercise. The starting point is still the same coordinate-space correlator,

$$\Pi^{ij}_{\mu\nu}(p) = -iN_c \int d^4x\, e^{ip\cdot x}\mathrm{Tr}\left[\Gamma^i_\mu S_c(-x)\Gamma^j_\nu S_b(x)\right], \tag{135}$$

but after the Dirac trace one keeps the Fourier transform of each Bessel product directly. A typical scalar integral has the form

$$\mathcal{I}^{(r)}_{n\mu\nu}(p^2) = \int d^4x\, e^{ip\cdot x}(p\cdot x)^r (x^2)^n K_\mu(m_c\sqrt{-x^2})K_\nu(m_b\sqrt{-x^2}). \tag{136}$$

The powers of $p\cdot x$ may be generated by differentiating the scalar Fourier integral with respect to p_α,

$$x_\alpha e^{ip\cdot x} = \frac{1}{i}\frac{\partial}{\partial p^\alpha}e^{ip\cdot x}, \tag{137}$$

or by doing Lorentz tensor reduction before the radial integral. The angular integration then gives integrals containing

$$J_1(Qx)K_\mu(m_c x)K_\nu(m_b x), \qquad Q^2 = -p^2. \tag{138}$$

At this stage the direct method must extract the discontinuity in $p^2 = s$ from the Bessel product itself. If Schwinger parameters are introduced only as a calculational aid, one meets distributions of the type

$$\delta\left(s - \frac{m_b^2}{x} - \frac{m_c^2}{1-x}\right), \tag{139}$$

and, when numerator powers or derivatives act on the Fourier kernel, derivatives of the same delta function,

$$\delta^{(n)}(s - \bar{s}(x)), \qquad \bar{s}(x) = \frac{m_b^2}{x} + \frac{m_c^2}{1-x}. \tag{140}$$

These objects must be treated analytically. For example,

$$\int_0^1 dx\, F(x)\, \delta(s - \bar{s}(x)) = \sum_{x_i} \frac{F(x_i)}{|\bar{s}'(x_i)|}, \qquad \bar{s}(x_i) = s, \tag{141}$$

where the roots x_i exist only if $s \geq (m_b + m_c)^2$. This condition is the same physical threshold and may be written as a Heaviside factor,

$$\Theta\big(s - (m_b + m_c)^2\big). \tag{142}$$

For derivative delta functions one must integrate by parts,

$$\int dx\, F(x)\, \delta^{(n)}(s - \bar{s}(x)), \tag{143}$$

carefully keeping the Jacobian factors and verifying that no endpoint terms survive after the physical support restriction is imposed.

14.3.1 Explicit treatment of δ and δ' terms

For the direct non-Azizi route we use the parameter assignment

$$\bar{s}(x) = \frac{m_b^2}{x} + \frac{m_c^2}{1-x} = \frac{m_c^2 x + m_b^2(1-x)}{x(1-x)}. \tag{144}$$

The physical support is obtained by solving $\bar{s}(x) < s$. The two roots are

$$x_\pm(s) = \frac{s + m_b^2 - m_c^2 \pm \sqrt{\lambda(s, m_b, m_c)}}{2s}, \tag{145}$$

and they are real only when

$$s \geq s_{\text{th}} = (m_b + m_c)^2. \tag{146}$$

The derivative needed in the Jacobian is

$$\bar{s}'(x) = -\frac{m_b^2}{x^2} + \frac{m_c^2}{(1-x)^2}. \tag{147}$$

Therefore the ordinary delta-function integral is

$$\begin{aligned} I_0[F](s) &= \int_0^1 dx\, F(x)\, \delta(s - \bar{s}(x)) \\ &= \Theta(s - s_{\text{th}}) \sum_{\eta=\pm} \frac{F(x_\eta(s))}{|\bar{s}'(x_\eta(s))|}. \end{aligned} \tag{148}$$

This is the explicit origin of both the threshold factor and the finite domain $x_-(s) < x < x_+(s)$.

For higher derivatives the cleanest identity is

$$\delta^{(n)}(s-\bar{s}(x)) = \frac{\partial^n}{\partial s^n}\delta(s-\bar{s}(x))\,, \tag{149}$$

where the derivative is with respect to the full argument $s-\bar{s}(x)$. Hence, at the level of a spectral density,

$$I_n[F](s) = \int_0^1 dx\, F(x)\,\delta^{(n)}(s-\bar{s}(x)) = \frac{d^n}{ds^n} I_0[F](s). \tag{150}$$

This formula is correct, but differentiating the explicit $\Theta(s-s_{\rm th})$ form can produce threshold-localized distributions. For the Borel sum rule it is usually safer to leave the delta derivative inside the s-integral until after continuum subtraction.

The continuum-subtracted Borel object for a generic term is

$$\mathcal{B}_n[W] = \int_0^1 dx \int_{s_{\rm th}}^{s_0} ds\, e^{-s/M^2} W(s,x)\delta^{(n)}(s-\bar{s}(x))\,. \tag{151}$$

Using the distribution identity

$$\int ds\,\phi(s)\delta^{(n)}(s-a) = (-1)^n\phi^{(n)}(a), \tag{152}$$

one obtains

$$\mathcal{B}_n[W] = \int_{x_-(s_0)}^{x_+(s_0)} dx\,(-1)^n \left[\frac{\partial^n}{\partial s^n}\left(e^{-s/M^2}W(s,x)\right)\right]_{s=\bar{s}(x)}. \tag{153}$$

This equation is the practical rule we should use for direct G^2 and G^3 terms. The finite limits $x_\pm(s_0)$ encode the condition $\bar{s}(x) < s_0$; the lower threshold is automatic because $\bar{s}(x) \geq (m_b+m_c)^2$ for $0<x<1$.

For the common case in which W has no explicit s-dependence,

$$\mathcal{B}_n[F] = \int_{x_-(s_0)}^{x_+(s_0)} dx\,\frac{F(x)}{(M^2)^n}\exp\left[-\frac{\bar{s}(x)}{M^2}\right]. \tag{154}$$

For the first two derivative cases with an explicitly s-dependent coefficient $W(s,x)$, the formulas are

$$\mathcal{B}_1[W] = \int_{x_-(s_0)}^{x_+(s_0)} dx\, e^{-\bar{s}(x)/M^2}\left[\frac{W(\bar{s}(x),x)}{M^2} - W_s(\bar{s}(x),x)\right], \tag{155}$$

$$\mathcal{B}_2[W] = \int_{x_-(s_0)}^{x_+(s_0)} dx\, e^{-\bar{s}(x)/M^2}\left[W_{ss}(\bar{s}(x),x) - \frac{2W_s(\bar{s}(x),x)}{M^2} + \frac{W(\bar{s}(x),x)}{(M^2)^2}\right]. \tag{156}$$

If a derivation instead produces $\delta^{(n)}(\bar{s}(x)-s)$, one must multiply the above result by $(-1)^n$, since

$$\delta^{(n)}(\bar{s}-s) = (-1)^n\delta^{(n)}(s-\bar{s}). \tag{157}$$

This sign convention must be fixed before coding any direct condensate term.

Thus the direct route has the schematic structure

$$\begin{aligned}&\text{Bessel product} \longrightarrow \text{radial Fourier transform} \longrightarrow \text{Disc}_{p^2=s}\\ &\longrightarrow \delta(s-\bar{s}),\ \delta'(s-\bar{s}),\ldots \longrightarrow \Theta(s-s_{\rm th}) \longrightarrow x_\pm(s).\end{aligned} \tag{158}$$

This is analytically possible, but it is precisely the bookkeeping that the Azizi et al. method systematizes. The direct section is therefore best used as a validation path: derive one channel, preferably AA, all the way to the same perturbative spectral density and confirm that the final answer is proportional to

$$\frac{\sqrt{\lambda(s, m_b, m_c)}}{s} \Theta\big(s - (m_b + m_c)^2\big) \,. \tag{159}$$

Only after the AA channel is reproduced should one repeat the calculation for AB and BB, because those channels additionally test the tensor current normalization $1/(m_b + m_c)$. The condensate terms are substantially more cumbersome in this route, since local G^2, local G^3, and the open-field $S_c^{(G)} S_b^{(G)}$ term can generate higher derivatives of the same support delta functions.

The file BcMixingCoordinateDirect.wl implements the first Mathematica version of this non-Azizi perturbative check. It follows the Huang–Liu/Groote–Körner–Pivovarov viewpoint rather than the Azizi Borel-continuum prescription. Here the discontinuity of two massive coordinate-space propagators is represented by the inverse-K-transform, equivalently the ordinary two-body cut. The code keeps the support structure explicit through

$$\bar{s}(x) = \frac{m_c^2 x + m_b^2(1-x)}{x(1-x)}, \qquad \delta(s - \bar{s}(x)) \longrightarrow \Theta\big(s - (m_b + m_c)^2\big) \,, \tag{160}$$

and records the corresponding roots $x_\pm(s)$. The entry points are

```
Get["BcMixingCoordinateDirect.wl"];
DirectDeltaSupportNote[]
DirectPerturbativeSpectralDensity["AA", s]
DirectPerturbativeCheck[8.0, 54.0]
```

After loading the ordinary coordinate-space file, the comparison

```
Get["BcMixingCoordinate.wl"];
DirectCompareToCoordinate[8.0, 54.0]
```

gives

```
<|"DirectPertDeg" -> 43.28944792785432,
  "CoordinatePertDeg" -> 43.28944792785432,
  "DifferenceDeg" -> 0.|>
```

Thus the direct non-Azizi perturbative discontinuity reproduces the same mixing angle at the Wang-window test point. This check should not yet be extended to condensates by guesswork. The G^2 and G^3 terms contain higher powers of the coordinate variable and therefore generate derivatives of the support delta functions; their direct treatment requires a separate analytic derivation.

A diagnostic condensate layer is also available in the direct-coordinate file. It does not claim to be a fresh coordinate-space derivation of the condensate spectral densities. Instead, it takes the pole expansion already obtained from the Feynman-parameter amplitude and rewrites each pole term as a $\delta^{(n)}(s - \bar{s})$ Borel weight. This is a controlled test of the distribution algebra. The entry points are

```
Get["BcMixingMomentum.wl"];
Get["BcMixingCoordinateDirect.wl"];
DirectCondensateWeightSummary["AA", "G2"]
DirectCondensateComparisonToMomentum[8.0, 54.0, "G2"]
DirectCondensateComparisonToMomentum[8.0, 54.0, "G3"]
```

For G^2, the derivative orders are $\{0,1,2\}$. For G^3, the derivative orders are $\{1,2,3,4\}$. At $M^2 = 8$ GeV^2, $s_0 = 54$ GeV^2, the direct derivative-delta reconstruction agrees with the corrected momentum direct-Borel moments:

$$\Pi_{G^2}^{AA} = -1.9315653037 \times 10^{-4}, \qquad \Pi_{G^3}^{AA} = 4.4587307978 \times 10^{-5}, \tag{161}$$

$$\Pi_{G^2}^{AB} = 1.2708857912 \times 10^{-4}, \qquad \Pi_{G^3}^{AB} = -3.4790161017 \times 10^{-5}, \tag{162}$$

$$\Pi_{G^2}^{BB} = -1.1094520542 \times 10^{-4}, \qquad \Pi_{G^3}^{BB} = 2.3373372969 \times 10^{-5}. \tag{163}$$

The agreement is exact at the displayed numerical precision. The code also now enforces that the explicit arguments of numerical routines, `M2` and `s0`, override any `"M2"` or `"s0"` entries stored in a parameter association. This prevents an old default Borel value from contaminating scans.

14.4 Where the Dirac delta and Heaviside functions enter

The reason the current Mathematica output does not visibly display δ-functions and Heaviside functions is not that they are absent from the coordinate-space method. Rather, in the current implementation they have already been integrated out or replaced by equivalent integration limits.

The coordinate-space product of two massive propagators generates scalar terms of the schematic form

$$T = \int d^4x\, e^{ip\cdot x} \frac{K_\nu(m_c\sqrt{-x^2})K_\mu(m_b\sqrt{-x^2})}{(\sqrt{-x^2})^n} \times \{\text{polynomial in } x^2, p\cdot x, p^2\}. \tag{164}$$

Using an integral representation for each modified Bessel function, one obtains Schwinger-parameter integrals of the schematic form

$$T \sim \int d\rho\, dv\, dw\, \rho^a v^b w^c (1-v-w)^d \exp\left[-\frac{m_c^2}{\rho v} - \frac{m_b^2}{\rho w} - \frac{Q^2}{4\rho}\right], \qquad Q^2 = -p^2. \tag{165}$$

This is the stage emphasized in the Azizi et al. treatment. In applications with two external virtualities, successive Borel transformations produce Dirac delta functions that fix combinations of the Schwinger parameters. A typical reduced structure is

$$\rho(s_1,s_2) \sim \int dz\, dy\, dl\, F(z,y,l)\, \delta\left[s_1 - l - \frac{m_c^2}{z(1-y)} - \frac{m_b^2}{zy}\right] \delta(s_2 - s_1), \tag{166}$$

up to derivatives of the delta functions when powers of the external invariants appear. After the l-integration, these delta functions turn into support restrictions,

$$\int_0^\infty dl\, F(l)\, \delta(s-l-A) = F(s-A)\, \Theta(s-A), \tag{167}$$

$$A = \frac{m_c^2}{z(1-y)} + \frac{m_b^2}{zy}. \tag{168}$$

Thus one obtains

$$\Theta\left(s_1 - \frac{m_c^2}{z(1-y)} - \frac{m_b^2}{zy}\right), \tag{169}$$

or equivalently into restricted integration limits. These Θ-functions are what remove apparent endpoint divergences in the Schwinger-parameter integrals.

For the present two-point mixing correlator there is one external invariant, $p^2 = s$, and the same physics appears in a simpler form: the spectral density has support only above

$$s_{\text{th}} = (m_b + m_c)^2, \tag{170}$$

or, in a Feynman-parameter representation,

$$s > \bar{s}(x) = \frac{m_c^2 x + m_b^2 (1-x)}{x(1-x)}. \tag{171}$$

Solving $\bar{s}(x) < s_0$ gives the finite integration interval

$$x_-(s_0) < x < x_+(s_0), \tag{172}$$

where

$$x_\pm(s_0) = \frac{s_0 + m_b^2 - m_c^2 \pm \sqrt{\lambda(s_0, m_b, m_c)}}{2s_0}. \tag{173}$$

This is the explicit replacement used in the numerical integrations:

$$\int_0^1 dx\, F(x)\, \Theta(s_0 - \bar{s}(x)) \quad \longrightarrow \quad \int_{x_-(s_0)}^{x_+(s_0)} dx\, F(x). \tag{174}$$

Equivalently, when writing the dispersion integral, the same support appears as

$$\int ds\, \rho(s) = \int ds\, \Theta\big(s - (m_b + m_c)^2\big)\, \rho(s), \tag{175}$$

with the continuum-subtracted Borel moment

$$\Pi(M^2, s_0) = \int_{(m_b+m_c)^2}^{s_0} ds\, e^{-s/M^2} \rho(s). \tag{176}$$

Thus the Heaviside support has not disappeared; it is encoded in the lower threshold and in the $x_\pm$ integration limits used by the numerical Borel integrals.

The present coordinate-space code has two layers:

1. For the perturbative check, it installs the final reduced spectral density proportional to $\sqrt{\lambda(s, m_b, m_c)}\, N^{ij}(s)/s$. Therefore no intermediate δ- or Θ-functions appear in the output.

2. For the local G^2 and G^3 checks, it uses the already-tested Feynman/Schwinger direct-Borel reduction helpers from `BcMixingMomentum.wl`. In this representation the support is again implemented by the integration limits $x_\pm(s_0)$, not by printing explicit Heaviside functions.

For a journal paper, it is therefore important to say explicitly that the Dirac delta functions and Heaviside functions occur at the Schwinger/Borel reduction stage, but in the numerical implementation they are already integrated out and appear as physical thresholds and finite parameter integration domains.

For a student checking this section by hand, the suggested coordinate-space exercise is:

1. Start from the AA current only. Insert $S_c^{(0)}(-x)$ and $S_b^{(0)}(x)$ and keep track of the sign of the $\not{x}$ term.

2. Evaluate the Dirac trace before projection. The trace should reduce to scalar combinations of x^2, p^2, and $p \cdot x$, multiplied by products of K_1 and K_2.

3. Apply the transverse projector and replace $p^2 \to s$, $x^2 \to x^2$, and $p \cdot x \to px$. This is what `CoordinateScalarize` does.

4. Identify the radial integral associated with each scalar term. Terms with powers of $p \cdot x$ can be generated by differentiating the Fourier factor with respect to p_μ, or equivalently by using tensor reduction before the angular integral.

5. Use the Bessel/Schwinger representation to obtain the physical cut in s. The final perturbative answer must be proportional to $\sqrt{\lambda(s, m_b, m_c)}$ and vanish at $s = (m_b + m_c)^2$.

6. Repeat the numerator check for AB and BB. These two channels are the best tests of the tensor-current normalization because each B current contributes one factor $1/(m_b + m_c)$.

The coordinate-space notebook separates these checks intentionally:

- first evaluate only `BcMixingCoordinate.wl`;
- record `CoordinateTraceKernel["AA","pert"]`;
- record `CoordinatePerturbativeSpectralDensity["AA",s]`;
- compute `CoordinateNumericMixingAngleDegrees[8,54,"pert"]`;
- only after this blind check, load `BcMixingMomentum.wl` and compare with

  `CoordinateCompareToMomentum[8,54]`

The Mathematica entry points are

```
Get["BcMixingCoordinate.wl"];
CoordinateCheckEnvironment[]
CoordinateTraceKernel["AA", "pert"] // Short
CoordinateNumericMixingAngleDegrees[8, 54, "pert"]
CoordinateWangWindowGrid[]
```

If the momentum-space file is also loaded, the two perturbative calculations can be compared by

```
Get["BcMixingMomentum.wl"];
Get["BcMixingCoordinate.wl"];
CoordinateCompareToMomentum[8, 54]
```

which gives

```
<|"CoordinatePertDeg" -> 43.28944792785432,
  "MomentumPertDeg" -> 43.28944792785432,
  "DifferenceDeg" -> 0.|>
```

for $M^2 = 8$ GeV^2 and $s_0 = 54$ GeV^2. This equality is a validation check only. It should be reported as an agreement between two independent perturbative reductions, not as an input to the coordinate-space calculation.

For the next stage, the coordinate-space G^2 and G^3 propagator kernels are already written as extension points. The local-condensate heavy-quark terms used in the code are

$$S_Q^{(G^2)}(x) = -\frac{\langle g_s^2 G^2\rangle}{2^6\, 3\, (2\pi)^2}\left[(im_Q\not{x} - 6)\frac{K_1(m_Q\sqrt{-x^2})}{\sqrt{-x^2}} + 4m_Q x^2 \frac{K_2(m_Q\sqrt{-x^2})}{-x^2}\right], \tag{177}$$

$$S_Q^{(G^3)}(x) = \frac{\langle g_s^3 G^3\rangle}{2^8\, 3^2\, (2\pi)^2}\left[-\frac{i\not{x}x^2}{m_Q}\frac{K_1(m_Q\sqrt{-x^2})}{\sqrt{-x^2}} + 4i\not{x}x^2\frac{K_2(m_Q\sqrt{-x^2})}{-x^2} + \frac{10x^4}{m_Q}\frac{K_2(m_Q\sqrt{-x^2})}{-x^2} + x^4\frac{K_1(m_Q\sqrt{-x^2})}{\sqrt{-x^2}}\right]. \tag{178}$$

The dimension–4 coordinate-space check must include

$$S_c^{(G^2)}(-x)S_b^{(0)}(x), \qquad S_c^{(0)}(-x)S_b^{(G^2)}(x), \tag{179}$$

and, if the open-field propagator is kept explicitly, the cross-line $S_c^{(G)}(-x)S_b^{(G)}(x)$ term. The present coordinate file records the local G^2 and G^3 kernels and can build their projected trace kernels, for example

```
CoordinateTraceKernel["AA", "G2c"]      (* S_c^(G2) S_b^(0) *)
CoordinateTraceKernel["AA", "G2b"]      (* S_c^(0)  S_b^(G2) *)
CoordinateTraceKernel["AA", "G2local"]  (* local sum of the two *)
CoordinateTraceKernel["AA", "G3local"]  (* local single-line G3 sum *)
```

At this stage, however, the numerical coordinate-space spectral/Borel integration is enabled independently only for `"pert"`. A reduced Bessel/Schwinger integration bridge exists for the local single-line condensate pieces, but that bridge is diagnostic: it is used only after loading `BcMixingMomentum.wl`, because that file contains the tested Feynman/Schwinger parameter reduction and direct Borel transform. Therefore these cells are not part of a coordinate-space paper-final OPE result:

```
Get["BcMixingMomentum.wl"];
Get["BcMixingCoordinate.wl"];

CoordinateLocalCondensateSummary[8, 54]
CoordinateNumericMixingAngleDegrees[8, 54, "pertG2local"]
CoordinateNumericMixingAngleDegrees[8, 54, "totalLocal"]
CoordinateLocalCondensateComparisonToMomentum[8, 54]
CoordinateLocalCondensateComparisonToMomentum[8, 54,
  $BcCoordinateDefaultParameters, "IncludeG3" -> True]
```

The labels mean

$$G2c : S_c^{(G^2)}(-x)S_b^{(0)}(x), \qquad G2b : S_c^{(0)}(-x)S_b^{(G^2)}(x), \tag{180}$$

$$G3c : S_c^{(G^3)}(-x)S_b^{(0)}(x), \qquad G3b : S_c^{(0)}(-x)S_b^{(G^3)}(x). \tag{181}$$

At $M^2 = 8$ GeV^2, $s_0 = 54$ GeV^2, the local-coordinate test gives

$$\theta_{\text{pert+G2local}} = 43.302779779^\circ, \qquad \theta_{\text{pert+G2local+G3local}} = 43.295658148^\circ. \tag{182}$$

The word "local" is important. It means that the condensate is inserted on one heavy-quark line at a time. For G^3, this reproduces the single-line momentum-space G^3 implementation used here. For G^2, it does not yet reproduce the complete dimension–4 result, because there is also an open-field term in which one background gluon is attached to the charm line and one to the bottom line.

Numerically, for the AA channel at the same test point,

$$\Pi^{AA}_{G^2,\text{local}} = -1.7043477947 \times 10^{-4}, \tag{183}$$

$$\Pi^{AA}_{G^2,\text{full momentum}} = -1.9315653037 \times 10^{-4}, \tag{184}$$

$$\Pi^{AA}_{G^2,\text{residual}} = -2.2721750899 \times 10^{-5}. \tag{185}$$

This residual is the target for the missing coordinate-space $S_c^{(G)}(-x)S_b^{(G)}(x)$ derivation. By contrast, the local coordinate and momentum-space G^3 moments agree numerically at this point:

$$\Pi^{AA}_{G^3,\text{local}} = 4.4587307978 \times 10^{-5}, \tag{186}$$

$$\Pi^{AA}_{G^3,\text{momentum}} = 4.4587307978 \times 10^{-5}. \tag{187}$$

This should not yet be quoted as the complete condensate-corrected coordinate result. The reason is that the full dimension–4 contribution also contains the cross-line $S_c^{(G)}(-x)S_b^{(G)}(x)$ term. The helper `CoordinateLocalCondensateComparisonToMomentum` reports the difference between the full momentum-space G^2 moment and the local coordinate $G2c+G2b$ moment; this residual is the piece that must be reproduced from the open-field coordinate-space propagator. The G^3 comparison is optional because it is slower; setting `"IncludeG3" -> True` checks that the local coordinate $G3c+G3b$ integration matches the momentum-space G^3 contribution.

14.5 What remains for an independent coordinate-space OPE

For a separate coordinate-space publication, the following objects must be derived without calling the momentum-space OPE engine:

1. The open-field coordinate-space propagator $S_Q^{(G)}(x)$, before vacuum averaging.
2. The cross-line trace
$$\mathrm{Tr}\Big[\Gamma_i S_c^{(G)}(-x)\Gamma_j S_b^{(G)}(x)\Big],$$
for $ij = AA, AB, BB$, including the tensor-current normalization $p^\alpha/(m_b+m_c)$.
3. The coordinate-space vacuum contraction
$$\left\langle g_s^2 G_A^{\alpha\beta} G_B^{\rho\sigma}\right\rangle = \frac{\delta_{AB}}{96}\langle g_s^2 G^2\rangle \left(g^{\alpha\rho}g^{\beta\sigma} - g^{\alpha\sigma}g^{\beta\rho}\right).$$
4. The Bessel/radial reduction of the resulting coordinate-space kernels. This may be done with the Huang–Liu/Azizi style Feynman-parameter representation, or with the direct delta-derivative support method, but the reduction must be self-contained in the coordinate-space derivation.
5. The independent implementation of
$$G2gg, \qquad G2full = G2c + G2b + G2gg, \qquad pertG2full = pert + G2full.$$
6. A decision on the dimension–6 sector: either justify the single-line G^3 truncation for the chosen OPE accuracy, or derive the corresponding open-field coordinate-space G^3 terms. Only then should the order `totalFull` be used for final uncertainty analysis.

Until these steps are complete, the coordinate-space uncertainty analysis should be quoted only for `"pert"`, e.g.

```
CoordinateMonteCarloMixingAngleUncertainty[
  500, $BcCoordinateDefaultUncertaintyRanges, "pert"]
```

The entries `"pertG2local"` and `"totalLocal"` are useful diagnostics for the size of local condensate insertions, but they are not independent full-OPE coordinate-space results.

The independent condensate derivation has therefore been moved to the separate workbench `BcMixingCoordinateOPE.wl` and notebook `BcMixingCoordinateOPE.nb`. These files do not load `BcMixingMomentum.wl`. Their role is to expose the coordinate-space kernels and the required Bessel/Schwinger reduction steps:

```
Get["BcMixingCoordinateOPE.wl"];
CoordinateOPEStatus[] // Dataset
CoordinateOPEG2Status["AA"] // Dataset
CoordinateOPEG2KernelReport["AA"] // Short
CoordinateOPEReductionPlan["G2c"] // Dataset
```

```
CoordinateOPEKernelInventory["AA", "G2c"] // Dataset
CoordinateOPEKernel["AA", "G2c"] // Short
CoordinateOPEG2ggOpenFieldPropagatorNote[] // Dataset
CoordinateOPEG2ggContraction[]
CoordinateOPEKernelInventory["AA", "G2gg"] // Dataset
CoordinateOPEKernelInventory["AA", "G2full"] // Dataset
CoordinateOPEKernelReductionTable["AA", "G2full"] // Dataset
CoordinateOPEKernelBorelTemplate["AA", "G2full"] // Short
CoordinateOPEExportG2KernelsTeX["AA"]
```

At the kernel level the coordinate-space G^2 sector now contains

$$G2full = G2c + G2b + G2gg,$$

where $G2gg$ is the open-field cross-line contribution obtained from

$$S_c^{(G)}(-x)S_b^{(G)}(x)$$

and the vacuum contraction of the two background field strengths. The independent Bessel/Schwinger reduction of all dimension–4 G^2 kernels is now implemented through the same pole-weight method. The workbench decomposes each projected kernel into terms of the form

$$C_i\,(p\cdot x)^{a_i}(x^2)^{b_i}r^{c_i}K_{\nu_i}(m_c r)K_{\mu_i}(m_b r), \qquad r=\sqrt{-x^2},$$

and attaches a formal Borel template containing the family of $\delta^{(n)}(s-\bar{s}(x))$ terms.

For each of $G2c$, $G2b$, and $G2gg$, this reduction is performed directly in `BcMixingCoordinateOPE.wl`. We define the pole weights by expanding each reduced term around the physical support,

$$\Pi^{ij}_{O,i}(s,x)=\sum_{r\geq 1}\frac{C^{ij}_{i,r}(x)}{[s-\bar{s}(x)]^r}, \qquad \bar{s}(x)=\frac{m_b^2}{x}+\frac{m_c^2}{1-x}.$$

The corresponding Borel weight used in the numerical integration is

$$\mathcal{B}^{ij}_{i,r}(x,M^2)=\frac{(-1)^r}{(r-1)!\,(M^2)^{r-1}}\,C^{ij}_{i,r}(x)\,\exp\left[-\frac{\bar{s}(x)}{M^2}\right].$$

The code check

$$\sum_{i,r}\mathcal{B}^{ij}_{i,r}(x,M^2)-\mathcal{B}^{ij}_{O}(x,M^2)=0, \qquad O\in\{G2c, G2b, G2gg\},$$

returns identically zero for AA, AB, and BB. This check is available as

```
CoordinateOPEG2PoleWeightCheck @@@
  Tuples[{{"AA", "AB", "BB"}, {"G2c", "G2b", "G2gg"}}]
```

which returns nine zeros.

Writing $\bar{x}=1-x$ and

$$A(x)=m_b^2\bar{x}+m_c^2 x,$$

the nonzero $G2gg$ pole weights are as follows. For the AA channel,

i	r	$C^{AA}_{i,r}(x)$
1	2	$-\dfrac{G_2A(x)}{144\pi^2 x\bar{x}}$
2	1	$-\dfrac{5G_2}{288\pi^2}$
2	2	$\dfrac{5G_2A(x)}{288\pi^2 x\bar{x}}$

For the AB channel,

i	r	$C_{i,r}^{AB}(x)$
1	1	$-\dfrac{G_2 m_c}{96(m_b+m_c)\pi^2\bar{x}}$
1	2	$-\dfrac{G_2 m_c A(x)}{96(m_b+m_c)\pi^2\bar{x}^2 x}$
2	1	$\dfrac{G_2 m_b}{96(m_b+m_c)\pi^2 x}$
2	2	$\dfrac{G_2 m_b A(x)}{96(m_b+m_c)\pi^2\bar{x}x^2}$

For the BB channel,

i	r	$C_{i,r}^{BB}(x)$
1	1	$-\dfrac{G_2 m_b m_c}{48(m_b+m_c)^2\pi^2\bar{x}x}$
1	2	$-\dfrac{G_2 m_b m_c A(x)}{48(m_b+m_c)^2\pi^2\bar{x}^2 x^2}$
2	1	$\dfrac{G_2 A(x)}{72(m_b+m_c)^2\pi^2\bar{x}x}$
2	2	$\dfrac{G_2 A(x)^2}{72(m_b+m_c)^2\pi^2\bar{x}^2 x^2}$
3	2	$-\dfrac{G_2 A(x)^2}{288(m_b+m_c)^2\pi^2\bar{x}^2 x^2}$

These are the explicit coordinate-space $W_{i,n}(x)$-type weights for the open-field $G2gg$ term in the pole convention above. The single-line weights $G2c$ and $G2b$ are generated by the same function `CoordinateOPEG2PoleWeights[channel, order]` and are checked by the same zero-residual test. Thus the full dimension–4 coordinate-space combination

$$G2full = G2c + G2b + G2gg$$

is now numerically available in the independent coordinate-space workbench.

At the central numerical point $M^2 = 8.0$ GeV^2 and $s_0 = 54.0$ GeV^2, the independent dimension–4 Borel moments are

	$G2c$	$G2b$	$G2gg$	$G2full$
AA	$1.16680219 \times 10^{-3}$	$2.77262067 \times 10^{-4}$	$5.15802955 \times 10^{-5}$	$1.49564455 \times 10^{-3}$
AB	$1.10755754 \times 10^{-3}$	$-3.36506716 \times 10^{-4}$	$4.26195485 \times 10^{-6}$	$7.75312778 \times 10^{-4}$
BB	$1.37364426 \times 10^{-3}$	$3.15779799 \times 10^{-4}$	$-3.22250025 \times 10^{-5}$	$1.65719906 \times 10^{-3}$

The full $G2full$ correction shifts the perturbative coordinate-space mixing angle from

$$\theta_{\rm pert} = 43.28944793^\circ \quad \text{to} \quad \theta_{\rm pert+G2full} = 43.31873333^\circ,$$

namely

$$\Delta\theta_{G2full} = +0.02928540^\circ.$$

The open-field part alone gives $\theta_{\rm pert+G2gg} = 43.25439835^\circ$, i.e. $\Delta\theta_{G2gg} = -0.03504958^\circ$. Before these coordinate-space G^2 numbers are used as paper-final, the overall sign and normalization convention of the coordinate-space Schwinger/Fourier phase should still be checked by hand against one channel.

The single-line dimension–6 contribution is implemented as

$$G3full = G3c + G3b,$$

with no open-field cross-line dimension–6 terms included. At the same central point,

	$G3c$	$G3b$	$G3full$
AA	$5.32974195 \times 10^{-4}$	$2.18319388 \times 10^{-5}$	$5.54806134 \times 10^{-4}$
AB	$4.80354418 \times 10^{-4}$	$-1.89557393 \times 10^{-5}$	$4.61398678 \times 10^{-4}$
BB	$3.87388600 \times 10^{-4}$	$1.55681046 \times 10^{-5}$	$4.02956705 \times 10^{-4}$

The single-line G^3 correction changes the angle from

$$\theta_{\text{pert+G2full}} = 43.31873333^\circ \quad \text{to} \quad \theta_{\text{totalFull}} = 43.22988065^\circ,$$

so

$$\Delta\theta_{G3} = -0.08885268^\circ.$$

At this point G^2/pert is roughly 2%–4%, while G^3/pert is roughly 1% in the three projected channels.

The current workbench also records the bookkeeping needed for that reduction. For each Bessel factor one uses the canonical Schwinger identity for $K_\nu(mr)/r^\nu$. Therefore a term with total Bessel order $\nu_i + \mu_i$ requires the canonical radial power $-(\nu_i + \mu_i)$. Any leftover power

$$c_i + (\nu_i + \mu_i)$$

is shown explicitly. If this residual is a non-negative even integer it can be generated by differentiating the Gaussian Fourier integral with respect to the Gaussian scale. If it is not, the term is flagged for a separate radial reduction or integration-by-parts step before numerical use. This is the point where most sign and dimension mistakes are expected, so it is intentionally kept visible in the notebook output.

The Gaussian derivative bookkeeping is checked in two independent ways. The working pole weights use ordinary Mathematica differentiation of the scalar source representation

$$A^{-2} \exp\left[-\frac{J^2}{4A}\right],$$

where $(p \cdot x)^a$ is generated by $(1/i)^a (p \cdot \partial_J)^a$ and $(x^2)^b$ by b scale or source-Laplacian insertions. The file `BcMixingCoordinateOPE.wl` now also contains the FeynCalc audit

```
CoordinateOPEFeynCalcGaussianMultiplierReport[]
```

which recomputes the same multipliers with `FourDivergence` and `FourLaplacian`. For the power pairs used in the dimension–4 and single-line dimension–6 coordinate-space kernels, the reported difference between the Mathematica-`D` multiplier and the FeynCalc multiplier is zero after setting $D = 4$. This check verifies the Lorentz derivative algebra; it is separate from the remaining global coordinate-space phase, sign, and normalization audit.

The order of completion should be

$$G2c, \quad G2b, \quad G2gg, \quad G2full = G2c + G2b + G2gg, \quad pertG2full = pert + G2full.$$

The workbench performs this bookkeeping for the three channels needed in the mixing-angle extraction,

$$AA, \qquad AB, \qquad BB,$$

through `CoordinateOPEG2CompletionReport[]` and `CoordinateOPEG2ReductionWorkbook[]`. The canonical dimension–4 and single-line dimension–6 condensate weights are now generated directly by

```
CoordinateOPEG2PoleWeights[channel, order]
CoordinateOPEG3PoleWeights[channel, order]
```

and checked through the zero-residual pole-weight tests. The manual `CoordinateOPEInstallWeight` path is retained only for future noncanonical kernels. The public numerical functions

```
CoordinateOPEBorelMoment[channel, order, M2, s0]
CoordinateOPEMixingAngle[M2, s0, order]
```

therefore evaluate `pert`, `G2full`, `pertG2full`, `G3full`, and `totalFull` directly in the coordinate-space workbench. The remaining physics task is to audit the shared sign and normalization convention and to decide whether the single-line G^3 truncation is sufficient, or whether open-field dimension–6 terms must be derived separately.

14.6 Coordinate-space OPE hierarchy scan

The coordinate-space OPE hierarchy is scanned with

```
coordinateOPEConvergence =
  CoordinateOPEConvergenceScan[{7.0, 9.0, 3}, {53.0, 55.0, 3}];

CoordinateOPEExportConvergenceCSV[
  coordinateOPEConvergence,
  "BcMixingCoordinateOPEConvergence_WangWindow.csv"]
```

The scan covers $M^2 = \{7, 8, 9\}$ GeV^2 and $s_0 = \{53, 54, 55\}$ GeV^2. The output table contains the signed channel ratios

$$\frac{\Pi^{ij}_{G^2}}{\Pi^{ij}_{\rm pert}}, \qquad \frac{\Pi^{ij}_{G^3}}{\Pi^{ij}_{\rm pert}}, \qquad ij = AA, AB, BB,$$

their maximum absolute values over the three channels, and the corresponding mixing angles $\theta_{\rm pert}$, $\theta_{\rm pert+G2full}$, and $\theta_{\rm totalFull}$.

The corresponding coordinate-space paper-facing decomposition plot is generated by

```
coordinateAziziThetaOrderM2 =
  CoordinateOPEMixingAngleOrderM2PublicationPlot[
    $BcCoordinateWangWindow["M2Range"],
    53.0,
    {"pert", "pertG2full", "totalFull"},
    $BcCoordinateDefaultParameters,
    "NPoints" -> 25,
    "YHalfWidth" -> 1.0
  ];
```

and the coordinate-space OPE hierarchy diagnostic is

```
coordinateAziziRatioM2 =
  CoordinateOPEContributionRatioM2PublicationPlot[
    $BcCoordinateWangWindow["M2Range"],
    53.0,
    $BcCoordinateDefaultParameters,
    "NPoints" -> 25,
    "UseAbs" -> True
  ];
```

The latter plots $|\Pi^{ij}_{G^2}/\Pi^{ij}_{\rm pert}|$ and $|\Pi^{ij}_{G^3}/\Pi^{ij}_{\rm pert}|$ for $ij = AA, AB, BB$. The G^3 entry is the single-line $G3c + G3b$ contribution; no open-field dimension–6 cross-line terms are included. The older grouped bar-chart helper,

```
CoordinateOPEContributionBarChart[
  m2Test, s0Test,
  $BcCoordinateDefaultParameters,
  "Normalization" -> "OverPert"]
```

is kept only for quick checks, not as the preferred paper plot.

At the central Wang-window point,

$$M^2 = 8.0\ \text{GeV}^2, \qquad s_0 = 54.0\ \text{GeV}^2,$$

the maximum absolute channel ratios are

$$\max_{ij}\left|\frac{\Pi^{ij}_{G^2}}{\Pi^{ij}_{\text{pert}}}\right| = 0.0384560963, \qquad \max_{ij}\left|\frac{\Pi^{ij}_{G^3}}{\Pi^{ij}_{\text{pert}}}\right| = 0.0134375395.$$

Thus the independent coordinate-space condensate corrections are smaller than the perturbative contribution at the central point. The corresponding angles are

$$\theta_{\text{pert}} = 43.2894479279^\circ, \qquad \theta_{\text{pert+G2full}} = 43.3187333275^\circ, \qquad \theta_{\text{totalFull}} = 43.2298806499^\circ.$$

On the 3×3 Wang-window grid the largest observed ratios are

$$\max_{\text{grid,ij}}\left|\frac{\Pi^{ij}_{G^2}}{\Pi^{ij}_{\text{pert}}}\right| = 0.0856492483, \qquad \max_{\text{grid,ij}}\left|\frac{\Pi^{ij}_{G^3}}{\Pi^{ij}_{\text{pert}}}\right| = 0.0294634786.$$

Both ratios decrease as M^2 is increased from 7 to 9 GeV^2. This is the expected OPE hierarchy pattern for the tested window. Before quoting these numbers as paper-final, the global coordinate-space sign and normalization audit should still be completed, and the single-line G^3 truncation should be stated explicitly.

14.7 Independent direct-coordinate route

The file `BcMixingCoordinateDirect.wl` gives a second coordinate-space calculation of the perturbative part. It avoids the Azizi et al. Borel continuum machinery and instead uses the direct support structure

$$\delta(s - \bar{s}(x)) \to \Theta\big(s - (m_b + m_c)^2\big) \sum_{x=x_\pm(s)} \frac{1}{|\bar{s}'(x)|}.$$

The notebook `BcMixingCoordinateDirect.nb` loads only the direct file in its main workflow. Its Monte Carlo uncertainty uses `DirectMonteCarloMixingAngleUncertainty`, not the momentum-space Monte Carlo engine. As with the standard coordinate-space file, full G^2 and G^3 direct-coordinate OPE terms remain pending and should not be filled by momentum-space values.

15 Checklist for independent verification

A theorist can check the calculation in the following order:

1. Confirm the current convention, especially the explicit factor of i in J^B_μ, the normalization $p^\alpha/(m_b + m_c)$, and the definition of $\sigma_{\mu\nu}$.
2. Reproduce the three projected perturbative traces and verify the numerator functions $N^{AA}(s)$, $N^{AB}(s)$, and $N^{BB}(s)$.
3. Check the two-body phase-space normalization $N_c\sqrt{\lambda}/(16\pi^2 s)$ used in $\rho^{ij}_{\text{pert}}(s)$.

4. Verify the background-field propagator terms and the color factor in the $S^{(G)}S^{(G)}$ contribution.
5. Independently derive the direct Borel formula for the G^2 and G^3 terms, including the $-i$ phase convention.
6. Check whether the standard single-line G^3 propagator contribution is sufficient, or whether cross-line open-field G^3 terms must be added.
7. Check that $\Pi_{G^2}/\Pi_{\rm pert}$ and $\Pi_{G^3}/\Pi_{\rm pert}$ remain small in the accepted Borel window.
8. Use the K_{ij} model only to study how channel-dependent perturbative corrections propagate into the angle; do not interpret it as a probability distribution or a replacement for the NLO calculation.
9. The true AA coefficient is now independently derived and tested. Derive the AB and BB two-loop perturbative spectral densities separately before quoting an NLO mixing angle.
10. Determine the final working window using OPE convergence, Borel stability, and pole-dominance criteria.